

Optimal Siting and Sizing of Solar, Wind, and Battery Storage to Compensate for Nuclear Decommissioning in France

Data Investigation Report
Matthieu Hautsch, Anna-Sherazade Sylla

Project Overview

This project aims to determine the optimal portfolio of renewable energy resources (wind, solar, and storage) for France in 2030, accounting for the retirement of nuclear reactors.

Part 1: Optimal Portfolio Determination

Representative Days and Load/Generation Profiles

To reduce computational complexity, 12 representative days (one per month) will be selected for 2024 using the k-medoids clustering algorithm, so that representative days correspond to actual historical days. Both national load and generation profiles are needed at hourly resolution. These data are publicly available from RTE's *éCO2mix* platform [1].

Nuclear Fleet Adjustment

Reactors are assumed to have a 50-year operational lifetime; units commissioned before 1980 will therefore be retired by 2030, removing approximately 10 GW from the current fleet. The Flamanville 3 EPR (1,650 MW_e) will be added to the existing Flamanville 1–2 units. Reactor commissioning dates and operational status data are sourced from the IAEA's Power Reactor Information System (PRIS) [2].

Regional Energy Balance

To assess where additional renewable capacity is needed, we will compute the hourly energy surplus or deficit for each of France's 12 administrative regions. This requires regional hourly generation and consumption data, disaggregated by energy source, available from RTE's regional *éCO2mix* dataset [3].

Transmission Network

Inter-regional transmission is modeled by aggregating the main 400 kV and 225 kV lines between regions, with georeferenced line data sourced from the Open Data Réseaux Énergies platform [4]. Link lengths are approximated using the population barycenters of connected regions and converted to costs via a per-kilometer assumption (not done yet).

Renewable Capacity Factors

Regional hourly capacity factors by region for wind and solar are retrieved from the Renewables.ninja platform [5], which provides modeled time series based on reanalysis weather data (ERA5 / MERRA-2) and the existing installed capacity configuration.

Techno-Economic Parameters

Capital and operational cost assumptions for wind, solar PV, and battery storage are drawn from RTE's long-term energy scenarios study *Futurs énergétiques 2050* [6], which provides detailed 2024 cost of each technology. Battery storage is modeled as a simple state-of-charge system with a 4-hour discharge duration. Only utility-scale solar PV is considered.

Part 2: Optimal Siting of Wind and Solar

Given the regional capacity targets from Part 1, France is discretized into a regular grid and each cell is assessed for suitability using land-use data from the Copernicus Land Monitoring Service [7]. Cell-level capacity factors are derived from ERA5 hourly data [8] to enable precise intra-regional siting of wind and solar assets.

References

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- [8] H. Hersbach et al., *ERA5 Hourly Data on Single Levels from 1940 to Present*, Copernicus Climate Change Service (C3S) Climate Data Store, <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels>, accessed May 2026.